Part B · MA 7 Differential Calculus

Reviewer code: _____ **Date returned:** _____ **Part A returned on:** _____

Before you start

- **Open this only after you have returned Part A.** Your Part A answers stand as you returned them.
- This form lists **61 proposed changes** to outcomes in **17 cases** of MA 7. Read the Part B reviewer guide first.
- The proposals were generated automatically by the syllabus-checking software while it was being tested. They are in no particular order, and identical proposals are shown once. None of them is endorsed by anyone. Judge each on its merits.
- If you cannot judge a proposal, tick **Cannot judge** and say why.
- Fill in this form in Microsoft Word or on paper, or use the response sheet `MA7-part-b-responses.csv`.

Course context

- **Course record** (*Synthetic: written for this review*): MA 7 Differential Calculus · BSIE · year 1 · engineering. See R-CASES and R-CODES in the references.
- **Course description** (*Synthetic: written for this review*): "Limits, continuity, derivatives and their applications to engineering problems."
- **Curriculum placement** (*Transcribed: copied into the system, not confirmed by the institution*): The BSIE 2022-2023 curriculum transcribed in the system lists MA 7 DIFFERENTIAL CALCULUS (year 1, 1st semester, 3 lecture and 0 laboratory units, prerequisite cell blank in the source). See R-BSIE and R-CODES.
- **Program outcomes in this course's cases** (*Synthetic: written for this review*): general wording written for this review, not BSIE's official outcomes. Judge them as written. The BSIE outcomes transcribed in the system are in R-BSIE.
  - Design solutions for complex engineering problems that meet specified needs.
  - Communicate effectively with a range of audiences.
  - Apply knowledge of mathematics and science to solve complex engineering problems.
  - Function effectively as a member or leader of a team.
  - Communicate effectively on complex engineering activities with the engineering community and society.
  - Recognize the need for, and engage in, independent and lifelong learning.
- **Course learning outcomes that the ILOs in this course's cases build toward** (*Synthetic: written for this review*):
  - Apply differentiation rules to solve optimization problems in engineering.
  - Define limits of functions.
  - Define the limit of a function.
- **Expected graduate attributes** (*Transcribed: copied into the system, not confirmed by the institution*): the four names held in the system. No description of them is held (R-EGA).
  - Critical and Creative thinker
  - Competent Catholic Augustinian-Marian Professional (short name in the system: "Competent Professional")
  - Socially Responsive Steward
  - Transformative Lifelong learner

MA7-01 · CLO and program outcome

Program outcome (PO)

> Design solutions for complex engineering problems that meet specified needs.

Course learning outcome (CLO), as written

> Define the derivative of a function.

MA7-01-R1 · Proposed re-mapping

Keep the CLO text unchanged and map it to a different program outcome. The proposal does not say which.

Rating:

- ☐ Appropriate
- ☐ Appropriate only if a suitable program outcome is named
- ☐ Not appropriate
- ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Which program outcome should it be mapped to, if any? Name it from the program outcomes listed under Course context or from your program's own outcomes, or write "none suitable". _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO itself should be revised ☐ The CLO should be removed ☐ No suitable program outcome exists ☐ Other: _____

Note:

Click or tap here to write.

MA7-01-R2 · Proposed rewrite of the CLO

Proposed text

> Design solutions to engineering problems that meet specified needs by applying derivatives, such as determining the dimensions of a container that minimize material cost within given constraints.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-01-R3 · Proposed rewrite of the CLO

Proposed text

> Design solutions to engineering problems that meet specified needs by applying the derivative of a function.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-01-R4 · Proposed rewrite of the CLO

Proposed text

> Design a solution to an engineering problem that meets specified needs by applying derivatives, such as dimensioning a container of minimum material cost.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-02 · CLO and program outcome

Program outcome (PO)

> Communicate effectively with a range of audiences.

Course learning outcome (CLO), as written

> Evaluate limits of rational functions.

MA7-02-R1 · Proposed rewrite of the CLO

Proposed text

> Evaluate limits of rational functions and communicate the steps and results effectively in written and oral form to peers and non-specialist audiences.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-02-R2 · Proposed re-mapping

Keep the CLO text unchanged and map it to a different program outcome. The proposal does not say which.

Rating:

- ☐ Appropriate
- ☐ Appropriate only if a suitable program outcome is named
- ☐ Not appropriate
- ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Which program outcome should it be mapped to, if any? Name it from the program outcomes listed under Course context or from your program's own outcomes, or write "none suitable". _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO itself should be revised ☐ The CLO should be removed
☐ No suitable program outcome exists ☐ Other: _____

Note:

Click or tap here to write.

MA7-02-R3 · Proposed rewrite of the CLO

Proposed text

> Communicate the evaluation of limits of rational functions clearly to technical and non-technical audiences using correct notation, graphs and written explanations.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-03 · CLO and program outcome

Program outcome (PO)

> Design solutions for complex engineering problems that meet specified needs.

Course learning outcome (CLO), as written

> Compute the derivative of a polynomial using the power rule.

MA7-03-R1 · Proposed rewrite of the CLO

Proposed text

> Design solutions to engineering problems that meet specified needs by applying derivatives, such as determining dimensions that minimize cost under given constraints.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-03-R2 · Proposed rewrite of the CLO

Proposed text

> Design a solution to an engineering problem that meets specified needs by using derivatives, including the power rule, to model and optimize it.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-03-R3 · Proposed re-mapping

Keep the CLO text unchanged and map it to a different program outcome. The proposal does not say which.

Rating:

- ☐ Appropriate
- ☐ Appropriate only if a suitable program outcome is named
- ☐ Not appropriate
- ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Which program outcome should it be mapped to, if any? Name it from the program outcomes listed under Course context or from your program's own outcomes, or write "none suitable". _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO itself should be revised ☐ The CLO should be removed ☐ No suitable program outcome exists ☐ Other: _____

Note:

Click or tap here to write.

MA7-05 · CLO and program outcome

Program outcome (PO)

> Apply knowledge of mathematics and science to solve complex engineering problems.

Course learning outcome (CLO), as written

> Use integration techniques to compute areas and volumes that arise in engineering problems.

MA7-05-R1 · Proposed rewrite of the CLO

Proposed text

> Use derivatives to compute rates of change and optimal values that arise in engineering problems.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-05-R2 · Proposed rewrite of the CLO

Proposed text

> Use differentiation techniques to solve rate-of-change and optimization problems that arise in engineering.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-05-R3 · Proposed rewrite of the CLO

Proposed text

> Use differentiation techniques to compute rates of change and optimal values that arise in engineering problems.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-05-R4 · Proposed rewrite of the CLO

Proposed text

> Apply differentiation techniques to solve engineering problems involving rates of change and optimization.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-06 · CLO and program outcome

Program outcome (PO)

> Apply knowledge of mathematics and science to solve complex engineering problems.

Course learning outcome (CLO), as written

> Understand calculus.

MA7-06-R1 · Proposed rewrite of the CLO

Proposed text

> Apply limits and derivatives to solve engineering problems.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-06-R2 · Proposed rewrite of the CLO

Proposed text

> Apply the concepts of limits, continuity and derivatives to solve engineering problems.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-06-R3 · Proposed rewrite of the CLO

Proposed text

> Apply the concepts of limits and derivatives to solve engineering problems.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately

☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-07 · CLO and a list of program outcomes

Program outcomes (the whole list)

> 1. Apply knowledge of mathematics and science to solve complex engineering problems.
> 2. Design solutions for complex engineering problems that meet specified needs.
> 3. Function effectively as a member or leader of a team.
> 4. Communicate effectively on complex engineering activities with the engineering community and society.
> 5. Recognize the need for, and engage in, independent and lifelong learning.

Course learning outcome (CLO), as written

> Solve systems of linear equations using matrix methods.

MA7-07-R1 · Proposed rewrite of the CLO

Proposed text

> Apply matrix methods to solve systems of linear equations that model engineering problems.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-07-R2 · Proposed rewrite of the CLO

Proposed text

> Apply knowledge of mathematics, including limits and derivatives, to solve complex engineering problems such as rates of change and optimization.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-07-R3 · Proposed rewrite of the CLO

Proposed text

> Apply differentiation and matrix methods to solve engineering problems involving rates of change and systems of linear equations.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-08 · CLO and a list of program outcomes

Program outcomes (the whole list)

> 1. Apply knowledge of mathematics and science to solve complex engineering problems.
> 2. Design solutions for complex engineering problems that meet specified needs.
> 3. Function effectively as a member or leader of a team.
> 4. Communicate effectively on complex engineering activities with the engineering community and society.
> 5. Recognize the need for, and engage in, independent and lifelong learning.

Course learning outcome (CLO), as written

> Collaborate with a team to design a prototype that meets stated requirements.

MA7-08-R1 · Proposed rewrite of the CLO

Proposed text

> Collaborate with a team to design a solution to an engineering problem by applying derivatives to meet stated requirements.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-08-R2 · Proposed rewrite of the CLO

Proposed text

> Design, with a team, solutions to complex engineering problems that meet specified needs, using derivatives to model and optimize the design.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-09 · CLO and a list of program outcomes

Program outcomes (the whole list)

> 1. Apply knowledge of mathematics and science to solve complex engineering problems.
> 2. Design solutions for complex engineering problems that meet specified needs.
> 3. Function effectively as a member or leader of a team.
> 4. Communicate effectively on complex engineering activities with the engineering community and society.
> 5. Recognize the need for, and engage in, independent and lifelong learning.

Course learning outcome (CLO), as written

> Appreciate the history of ancient Rome and its art.

MA7-09-R1 · Proposed rewrite of the CLO

Proposed text

> Apply limits and differentiation rules to solve engineering problems.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-09-R2 · Proposed rewrite of the CLO

Proposed text

> Apply knowledge of limits, continuity and derivatives to solve complex engineering problems.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-09-R3 · Proposed rewrite of the CLO

Proposed text

> Apply differentiation techniques to solve engineering problems involving rates of change and optimization.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-09-R4 · Proposed rewrite of the CLO

Proposed text

> Apply derivatives to solve engineering problems involving rates of change and optimization.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-10 · CLO and expected graduate attribute

Expected graduate attribute (EGA)

> Socially Responsive Steward

Course learning outcome (CLO), as written

> Compute the derivative of trigonometric functions.

MA7-10-R1 · Proposed rewrite of the CLO

Proposed text

> Apply derivatives of trigonometric functions to model and solve engineering problems of community concern, such as vibration in structures, and explain the benefit of the solution to society.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-10-R2 · Proposed rewrite of the CLO

Proposed text

> Apply derivatives of trigonometric functions to model and analyze periodic engineering phenomena, such as vibration or electrical signals, and explain how the results serve community needs responsibly.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-10-R3 · Proposed re-mapping

Keep the CLO text unchanged and map it to a different graduate attribute. The proposal does not say which.

Rating:

- ☐ Appropriate
- ☐ Appropriate only if a suitable graduate attribute is named
- ☐ Not appropriate
- ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Which graduate attribute should it be mapped to, if any? Name one of the four graduate attributes listed under Course context, or write "none suitable". _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO itself should be revised ☐ The CLO should be removed ☐ No suitable graduate attribute exists ☐ Other: _____

Note:

Click or tap here to write.

MA7-11 · CLO and expected graduate attribute

Expected graduate attribute (EGA)

> Transformative Lifelong learner

Course learning outcome (CLO), as written

> Solve quadratic equations.

MA7-11-R1 · Proposed re-mapping

Keep the CLO text unchanged and map it to a different graduate attribute. The proposal does not say which.

Rating:

- ☐ Appropriate
- ☐ Appropriate only if a suitable graduate attribute is named
- ☐ Not appropriate
- ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Which graduate attribute should it be mapped to, if any? Name one of the four graduate attributes listed under Course context, or write "none suitable". _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO itself should be revised ☐ The CLO should be removed ☐ No suitable graduate attribute exists ☐ Other: _____

Note:

Click or tap here to write.

MA7-11-R2 · Proposed rewrite of the CLO

Proposed text

> Evaluate one's own understanding of limits and derivatives and apply independent study strategies to solve differentiation problems.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-11-R3 · Proposed rewrite of the CLO

Proposed text

> Apply derivatives to analyze engineering problems and reflect on gaps in one's own understanding to plan further independent study of calculus.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-11-R4 · Proposed rewrite of the CLO

Proposed text

> Evaluate one's own understanding of limits and derivatives and independently choose study strategies and resources to solve differential calculus problems.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-13 · CLO and expected graduate attribute

Expected graduate attribute (EGA)

> Critical and Creative thinker

Course learning outcome (CLO), as written

> Recite the multiplication table.

MA7-13-R1 · Proposed rewrite of the CLO

Proposed text

> Analyze a given engineering situation and select an appropriate derivative-based approach to solve it, justifying the choice.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-13-R2 · Proposed rewrite of the CLO

Proposed text

> Analyze calculus problems critically and propose creative solution strategies, justifying the choice of method for finding derivatives.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-13-R3 · Proposed rewrite of the CLO

Proposed text

> Analyze an engineering situation and justify the choice of differentiation technique used to solve it.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-13-R4 · Proposed rewrite of the CLO

Proposed text

> Analyze an engineering problem and justify the choice of differentiation technique used to solve it.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-14 · CLO and the list of graduate attributes

Expected graduate attributes (the whole list)

> 1. Critical and Creative thinker
> 2. Competent Catholic Augustinian-Marian Professional
> 3. Socially Responsive Steward
> 4. Transformative Lifelong learner

Course learning outcome (CLO), as written

> Recite the multiplication table.

MA7-14-R1 · Proposed rewrite of the CLO

Proposed text

> Analyze engineering problems critically and creatively, using limits and derivatives to build and justify solutions.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-14-R2 · Proposed rewrite of the CLO

Proposed text

> Analyze engineering problems and select appropriate differentiation techniques to solve them.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-14-R3 · Proposed rewrite of the CLO

Proposed text

> Analyze engineering problems and select appropriate differentiation methods to solve them, justifying the chosen approach.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-14-R4 · Proposed rewrite of the CLO

Proposed text

> Analyze engineering problems and devise creative solutions using limits and derivatives.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The CLO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-15 · ILO and course learning outcome

Course learning outcome (CLO)

> Apply differentiation rules to solve optimization problems in engineering.

Intended learning outcome (ILO), as written

> Differentiate polynomial and trigonometric functions.

MA7-15-R1 · Proposed rewrite of the ILO

Proposed text

> Apply differentiation rules to polynomial and trigonometric functions as a step toward setting up and solving simple optimization problems in engineering.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The ILO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-15-R2 · Proposed rewrite of the ILO

Proposed text

> Apply differentiation rules to polynomial and trigonometric functions to set up and solve simple engineering optimization problems.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The ILO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-15-R3 · Proposed rewrite of the ILO

Proposed text

> Apply differentiation rules to polynomial and trigonometric functions as a step toward setting up and solving simple engineering optimization problems.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The ILO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-15-R4 · Proposed rewrite of the ILO

Proposed text

> Apply differentiation rules to polynomial and trigonometric functions to solve simple optimization problems in engineering.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The ILO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-17 · ILO and course learning outcome

Course learning outcome (CLO)

> Apply differentiation rules to solve optimization problems in engineering.

Intended learning outcome (ILO), as written

> Sketch the graph of a given function.

MA7-17-R1 · Proposed rewrite of the ILO

Proposed text

> Sketch the graph of a given function by applying differentiation rules to locate its maxima and minima, as a step toward solving engineering optimization problems.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The ILO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-17-R2 · Proposed rewrite of the ILO

Proposed text

> Use the first derivative to sketch the graph of a given function and identify its maximum and minimum points.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The ILO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-17-R3 · Proposed rewrite of the ILO

Proposed text

> Apply the first derivative to sketch the graph of a given function and locate its maximum and minimum points for an engineering optimization problem.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The ILO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-17-R4 · Proposed rewrite of the ILO

Proposed text

> Sketch the graph of a given function using its first derivative to locate maxima and minima, as a step toward solving engineering optimization problems.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The ILO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-18 · ILO and course learning outcome

Course learning outcome (CLO)

> Define limits of functions.

Intended learning outcome (ILO), as written

> Design and implement a numerical solver for limits.

MA7-18-R1 · Proposed rewrite of the ILO

Proposed text

> Define the limit of a function using proper notation and describe it from a graph.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The ILO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-18-R2 · Proposed rewrite of the ILO

Proposed text

> State the definition of the limit of a function using correct notation.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The ILO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-18-R3 · Proposed rewrite of the ILO

Proposed text

> Define the limit of a function at a point.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The ILO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately

☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-18-R4 · Proposed rewrite of the ILO

Proposed text

> Define the limit of a function.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The ILO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-19 · ILO and course learning outcome

Course learning outcome (CLO)

> Apply differentiation rules to solve optimization problems in engineering.

Intended learning outcome (ILO), as written

> Describe the history of ancient Rome.

MA7-19-R1 · Proposed rewrite of the ILO

Proposed text

> Apply differentiation rules to find the maximum or minimum of a function that models an engineering quantity.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The ILO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-19-R2 · Proposed rewrite of the ILO

Proposed text

> Apply the power rule and product rule to differentiate functions that model engineering quantities.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The ILO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-19-R3 · Proposed rewrite of the ILO

Proposed text

> Apply the first-derivative test to find the maximum or minimum value of a function that models an engineering quantity.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The ILO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-19-R4 · Proposed rewrite of the ILO

Proposed text

> Apply derivatives to find the maximum or minimum value of a quantity in a simple engineering situation.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The ILO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-20 · ILO and course learning outcome

Course learning outcome (CLO)

> Define the limit of a function.

Intended learning outcome (ILO), as written

> Prove the epsilon-delta definition of a limit for arbitrary functions.

MA7-20-R1 · Proposed rewrite of the ILO

Proposed text

> State the definition of the limit of a function and explain it in terms of values approaching a number.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The ILO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-20-R2 · Proposed rewrite of the ILO

Proposed text

> State the definition of the limit of a function and explain it using the epsilon-delta idea for simple functions.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The ILO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-20-R3 · Proposed rewrite of the ILO

Proposed text

> State the definition of the limit of a function and explain it in words and with the epsilon-delta idea for simple functions.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The ILO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.

MA7-20-R4 · Proposed rewrite of the ILO

Proposed text

> State the definition of the limit of a function, including the epsilon-delta form, and explain its meaning.

Rating: ☐ Appropriate as written ☐ Acceptable with edits ☐ Not appropriate ☐ Cannot judge

If you cannot judge, why? ☐ Outside my expertise ☐ Insufficient information ☐ Other: _____

Concerns (tick any that apply): ☐ No concerns ☐ The ILO needed no change
☐ Raises the cognitive level more than the course or year warrants ☐ Lowers the cognitive level inappropriately
☐ Changes the course's subject matter or scope ☐ Duplicates or restates another outcome
☐ Adds, removes or changes values or attributes ☐ Adds content the alignment does not need
☐ Removes content the outcome needs ☐ A different remedy would be better (re-map, remove)
☐ Wording, clarity or measurability ☐ Other: _____

Suggested edit or note:

Click or tap here to write.